Discovering Android Things, an Embedded OS for IoT Devices

Android Things, an Android based embedded OS, has been designed for use with IoT devices. Here is a tutorial on connecting Android Things with Raspberry Pi to write a small, interesting program.

The Internet of Things has been expanding at a rapid rate over the past few years. The number of devices using the Internet to stay connected has also increased as we see a lot of smart devices all around us. Google has now made a move into the IoT market, by officially introducing the stable Android Things 1.0 in May 2018. The latest version of Android Things is 1.0.2, which was released in July 2018.

Android Things is an Android based embedded operating system, designed for use with the low-power, memory-constrained IoT devices. The main advantage of Android Things is that it uses the same Android development tools, such as Android Studio, along with Google APIs. So Android developers can use Java or Kotlin to write a program that can run an IoT device.

As always, Android keeps its products updated through regular releases, along with bug fixes, and also allows developers to release updates for the apps they developed through the Android Things console, similar to Android Apps updates in Google Play. Figure 1 shows the architecture of Android Things.

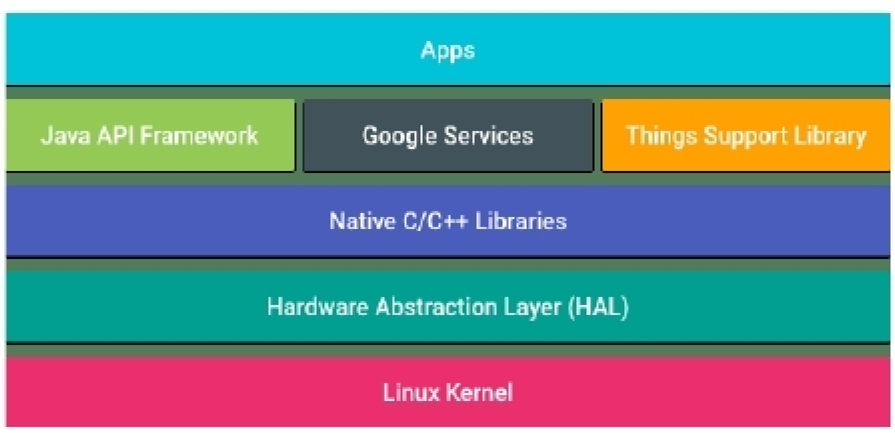

Figure 1: Android Things' architecture

Developing a program in Android Things that makes an LED blink

Hardware requirements:

- Raspberry Pi 3 Model B board
- One LED bulb, a resistor of 1K Ohm, jumper wires and a breadboard

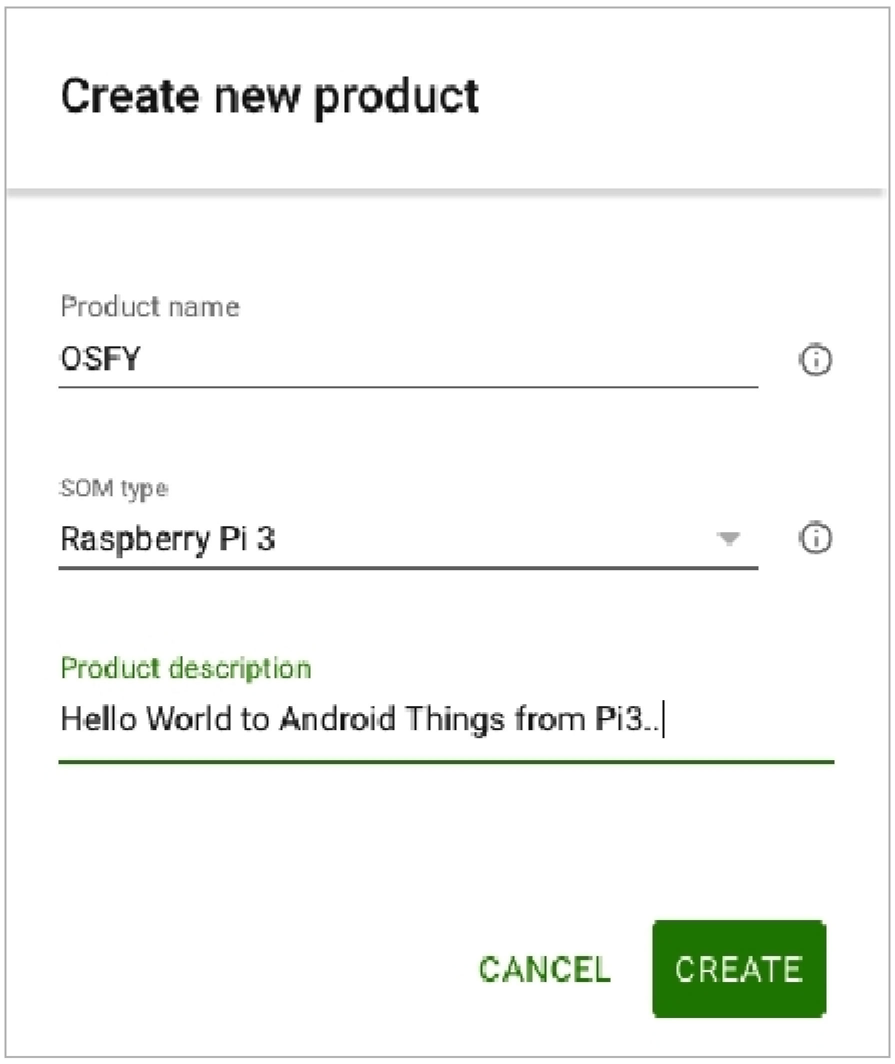

Figure 2: Creating a new product